Neeraj Varshney

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Research Summary

Senior Applied Scientist at Amazon driving foundation-model development across **pretraining, post-training (SFT and reinforcement learning)**, and **agentic systems**. Led data and modeling initiatives across three generations of **Mixture-of-Experts** models (trained on **23T+ tokens** and scaling to **800B+ parameters**), including a **web-scale data curation stack** that produced training data **outperforming leading public corpora** across knowledge, reasoning, and coding benchmarks. On the post-training side, owned the full development of **agentic models for tool use and retrieval planning** – from **synthetic data generation** through SFT, RLVR with **execution-grounded verifiable rewards** and **OPD**. **3,000+ citations**, publications at **ACL, EMNLP, NAACL, and EACL**, and multiple industry and academic research awards.

Technical Skills

Foundation Models : Pretraining, Mid-Training, Post-Training, Mixture-of-Experts, Distillation, Scaling Laws

Agents, Tool Use, & RL : Agentic Systems, Retrieval Planning, RLVR (GRPO/GSPO), Reward Design, Test-Time RL, On-Policy Distillation

Data & Evaluation : Web-Scale Data Curation, Quality Modeling, Synthetic Data, LLM-as-Judge, Deduplication, Taxonomy Design, Data-Mixture Optimization, Benchmarking

LLM Systems : PyTorch, Transformers/TRL, Megatron-LM, Slime, vLLM

Infrastructure : Python, PySpark, Pandas, AWS (Bedrock, S3, EC2), MLflow, Weights & Biases

Experience

Amazon Science 2024 – Present
Senior Applied Scientist *Palo Alto, California*

▷ **Foundation Models & Pretraining**

- Owned **pretraining data strategy and curation** across three generations of large-scale **Mixture-of-Experts foundation models**, spanning quality modeling, filtering, deduplication, taxonomy, and data-mixture optimization for models trained on **23T+ tokens** and scaling to **800B+ parameters**.
- Built a **web-scale data quality modeling stack** from LLM-based annotation to production-scale inference; quality-filtered data **outperformed leading public corpora including Nemotron-CC, FineWeb, and DCLM** across knowledge, reasoning, and coding benchmarks.
- Designed a **14-dimensional web-data taxonomy** covering orthogonal capability and quality signals including **reasoning depth**, **timeliness, and cultural specificity**; taxonomy-guided filtering recovered high-value data from lower-tier web corpora and enabled an **additional training stage without data repetition**.
- Led **data-mixture optimization strategies** across web, code, multilingual, and capability-targeted datasets, identifying **quality–composition–scale tradeoffs** that informed pretraining data allocation.

▷ **Reasoning, Synthetic Data & Capability Development**

- Improved **code capability** by distilling LLM code-quality judgments into a scalable filter — lifting HumanEval +9.64% and MBPP +8.21% on quality-selected code data.
- Drove **large-scale synthetic reasoning data generation**, transforming web and domain-specific documents into multi-step QA, knowledge-graph reasoning, and execution-grounded training data; generated **40B+ tokens** of reasoning-oriented data spanning web, code, and mathematics.
- Developed **code-reasoning** and **agentic training data** that model iterative problem solving through **failed attempts, execution feedback, correction, and verification**, extending static code data toward **state–action–observation trajectories**.

▷ **Post-Training, Reinforcement Learning & Agentic Systems**

- Led development of an **RL-trained retrieval planner** for Amazon's centralized customer-data platform (Customer360), translating natural-language customer queries into executable tool-call plans over a multi-resource retrieval API; trained a **Qwen3.5-4B** with **SFT + RLVR (GRPO)** where the reward is computed by executing both policy and teacher plans against the live service and scoring record-set overlap — reaching **0.67 retrieval F1 at 0.91 resource selection**; also **cutting serving output by 70%** to meet the production latency budget via a compact action format.

- Solved the **no-ground-truth problem** for retrieval supervision with an **"answer-first" synthetic-data pipeline** — a teacher LLM generates queries grounded in real customer records and cites the answering evidence, with a snapshot-blind plan teacher to prevent inference-time leakage — and built the execution-validation and RL-reward stack on top.
- Designed **Plan-DAG**, a structured planner that emits a dependency graph for multi-hop retrieval and runs exact set/aggregation operations in a deterministic, low-latency executor — shifting work from approximate generation to verifiable execution and reducing downstream retrieval payload up to **27×**.

Tencent AI May 2023 – Aug 2023
NLP Research Intern *Bellevue, Washington*

- Developed a **hallucination detection and mitigation framework** that identifies low-confidence generations, validates claims against external evidence, and repairs hallucinations during decoding; paper **"A Stitch in Time Saves Nine,"** is now widely cited.

Amazon Science May 2022 – Aug 2022
Applied Scientist Intern, Alexa AI *Seattle, Washington*

- Developed efficient **open-domain QA** systems combining parametric knowledge with retrieval-based reasoning.

Microsoft July 2018 – Aug 2019
Software Developer *Bangalore, India*

- Developed **ML-driven recommendation systems** for Microsoft Teams and built production services supporting **millions of enterprise users**.

Samsung R&D Institute Summer 2017
Research Intern *Bangalore, India*

- Developed a 'context prediction' application leveraging event features such as app usage, location, and sensor data.

Selected Papers

Full publication list: Google Scholar (3,000+ citations)

Foundation Models & Data-Centric AI

1. Unlocking Latent Value: Taxonomy-Guided Recovery of High-Performing Data from Low-Tier Web Corpora
Neeraj Varshney, Sanket Lokegaonkar, Nasser Zalmout, Qingyu Yin, Priyanka Nigam, Bing Yin

- Introduced a **taxonomy-guided web-data curation framework** with orthogonal quality dimensions and a distilled **0.5B / 73M-MLP annotator** achieving **50× throughput** across 14M documents.
- Demonstrated that taxonomy-filtered lower-tier web data can **outperform unfiltered top-tier corpora**, improving coding by up to **22.3%** and reasoning by up to **19.5%**.

2. Expert Upcycling: Shifting the Compute-Efficient Frontier of Mixture-of-Experts
Chaitanya Dwivedi, Binxuan Huang, Himanshu Gupta, Pratik Jayarao, Neeraj Varshney, Bing Yin

- Introduced Expert Upcycling, a method that grows **Mixture-of-Experts** capacity by duplicating high-utility experts and extending the router during continued pretraining – expanding model size while holding top-K routing (and per-token inference cost) fixed.
- Matched the from-scratch baseline's quality while **saving 32% of GPU hours** on 7B–13B models, and added gradient-based utility selection that **more than tripled quality-gap closure** under limited compute.

Reinforcement Learning & Reasoning

3. Translate-R1: Cost-Aware Translation Tool Use via Reinforcement Learning
Pratik Jayarao, Chaitanya Dwivedi, Himanshu Gupta, Neeraj Varshney, Adithya M Devraj, Meet Vadera, Priyanka Nigam, Bing Yin

- Trained a single RL policy that learns *when* to invoke translation as a tool – via **confidence-gated GSPO** for cost-sensitive tool use – on Qwen3-4B across **22 languages**, three resource tiers, and five domains, replacing hand-built routing rules with reward-driven introspection.
- Lifted reward by up to **+23.5 on low-resource tiers** while preserving full task reward at just **66% of the always-translate cost**, and generalized **zero-shot to 9 held-out languages**.

4. Toward Honest Language Models for Deductive Reasoning
Jiarui Liu, Kaustubh Dhole, Yingheng Wang, Haoyang Wen, Sarah Zhang, Haitao Mao, Gaotang Li, Neeraj Varshney, Jingguo Liu, Xiaoman Pan

- Studied **honest deductive reasoning** – models must answer only when a conclusion is logically entailed by the premises and abstain otherwise – and curated two graph-derived benchmarks (linear-algebra and logical-inference) with unanswerable cases.
- Introduced **ANCHOR**, an RL method that injects ground-truth trajectories into rollouts to prevent the early-training collapse that sinks standard GRPO when negative rewards dominate – stabilizing training and markedly improving reasoning accuracy and honest abstention.

LLM Reliability & Hallucination

5. A Stitch in Time Saves Nine: Detecting and Mitigating Hallucinations of LLMs by Validating Low-Confidence Generation

Neeraj Varshney, Wenlin Yao, Hongming Zhang, Jianshu Chen, Dong Yu

- Developed a three-stage method that **actively detects and mitigates hallucinations during generation** – flagging low-confidence spans via the model's logits, validating them against external evidence, and repairing them before continuing – across article generation, multi-hop QA, and false-premise QA.
- Cut GPT-3.5's hallucination rate from **47.5% to 14.5%** at ∼**88% detection recall**, generalized to a different model family (Vicuna), and introduced no new hallucinations on false positives.

6. Post-Abstention: Towards Reliably Re-Attempting the Abstained Instances in QA

Neeraj Varshney, Chitta Baral

- Introduced and mathematically formulated the **post-abstention** task – re-attempting the instances a selective-prediction system abstained on, to raise coverage without sacrificing accuracy.
- Developed RETOP, which re-examines the model's top-N predictions with an auxiliary verifier to recover correct answers, delivering **consistent risk reductions across all 11 QA datasets** (up to **24.2 aggregate risk improvement**) in both in-domain and OOD settings.

Efficient Inference & Model Cascading

7. Investigating Acceleration of LLaMA Inference by Enabling Intermediate Layer Decoding via Instruction Tuning with 'LITE'

Neeraj Varshney, Agneet Chatterjee, Mihir Parmar, Chitta Baral

- Introduced LITE, adding explicit losses from a model's intermediate layers during instruction tuning so earlier layers gain generation ability without degrading the final layer – enabling token-level, confidence-based early exits.
- Cut LLaMA-2 inference cost by **37.9% (7B)** and **46.4% (13B)** with no loss in generation quality across four instruction test sets.

Evaluation & Benchmarking

8. ILDAE: Instance-Level Difficulty Analysis of Evaluation Data

9. NumGLUE: A Suite of Fundamental yet Challenging Mathematical Reasoning Tasks

10. LogicBench: Towards Systematic Evaluation of Logical Reasoning Ability of Large Language Models

Books

Advances in Multimodal Information Retrieval and Generation Springer, 2024

Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, Chitta Baral
Synthesis Lectures on Computer Vision (SLCV)

- A comprehensive treatment of Transformer-based multimodal retrieval, generation, and retrieval-augmented generation across vision and language, covering foundational methods, architectures, and emerging research directions for students, researchers, and practitioners.

Education

Arizona State University Tempe, AZ, USA
Ph.D. in Computer Science *2019 – 2024*

- **Advisor:** Chitta Baral | **GPA:** 4.0/4.0 | **Outstanding CS PhD Graduating Student Award**, 2024

BITS Pilani, Pilani Campus, India Pilani, India
B.E (Hons) Computer Science *2014-2018*

- **GPA**: 9.11/10 (with Distinction)

Honors and Awards

- **Amazon Performance Award** – for high-impact contributions to foundation-model training data quality and downstream model performance
- **Outstanding CS PhD Graduating student Award**, Arizona State University, 2024
- **Outstanding Reviewer** for EACL'23 (Question Answering track).
- **Outstanding Research Award**, GPSA ASU, 2023
- SCAI **Doctoral Fellowship**, ASU, 2023.
- ASU **Jumpstart Research Grant**, 2023 and 2024.

Service

Reviewing & Editorial

- **Area Chair**, ACL Rolling Reviews
- **Reviewer**: ACL, EMNLP, EACL (Outstanding Reviewer), Computational Linguistics Journal, COLM, AMLC, ARR

Community & Outreach

- Author of 20+ ML/NLP articles on Medium with 100K+ cumulative views.
- Mentored several PhD interns and supported multiple co-authored publications.